

A tale of two EVENs

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Abstract

Although it is generally considered that *even* triggers both additive and scalar inferences, there are cases where only one of them can be detected. We argue that the two EVENs in Mandarin Chinese (*shenzhi* and *lian.dou*) may provide a clue to this puzzle. Specifically, we show that the clausal EVEN (*shenzhi*) always gives rise to a scalar inference, and its additive inference is canceled when the alternatives are mutually incompatible. Meanwhile, the phrasal EVEN (*lian.dou*) displays an almost opposite pattern, where the additive inference is always triggered, while the scalar one is sometimes less obvious. We also observe that the scalar inferences triggered by the two EVENs in Mandarin Chinese can involve different scales. The clausal *shenzhi* always calls for a scale that is based on a contextually supplied gradable property (Greenberg 2018). We propose that the clausal *shenzhi* in Mandarin Chinese is a speaker-oriented element, so that the scalar relation is always based on the speaker's evaluation of how different accessible possible worlds are compared with each other within a contextually supplied scale. On the other hand, we argue that the phrasal EVEN *lian.dou* has an inherent additive meaning, and it obtains the scalar meaning by being associated with an Emphatic Assertion operator (Krifka 1995a). Specifically, *lian* 'include' introduces an object to be included in the initial domain associated with the distributive/universal particle *dou*, and the Emphatic Assertion operator triggers an inference that the newly included object is most unlikely to be included in the domain. This creates a domain-widening effect that gives rise to the scalar inference. Finally, we support our proposal by examining cross-linguistic patterns of scalar additive particles.

Keywords: scalar inference; additive inference; syntax-semantics interface; *lian...dou* construction; *even* construction

1. Introduction

It has been observed that a focus-sensitive particle like *even* has two sides of meanings. The truth-conditional meaning is an assertion of the sentence-radical excluding the particle *even*, or the prejacent. The non-truth-conditional meaning contains two types of inferences: the additive inference and the scalar inference (Horn 1969; Karttunen and Peters 1979; Kay 1990; Krifka 1998a; Herburger 2000).¹ The additive inference suggests that an additional entity (which is comparable, but not identical, to the focused entity XP_F) also has the said property, while the scalar inference implies a scalar relation between the focused entity and its alternatives, so that among the alternative set, the focused entity is the most unlikely (or the most noteworthy) entity that has the said property. These meanings can be illustrated in (1):

¹ The additive and scalar meanings associated with *even* have been analyzed as either presuppositions (Horn 1969; Kay 1990) or conventional implicatures (Karttunen and Peters 1979), and it has also been argued that the additive meaning of German *auch nur* 'even_{NPI}' should be analyzed as an entailment (Schwarz 2005). Since our paper focuses on the lexical items in Mandarin Chinese that may bring about the meanings akin to *even* in English, rather than on the distinction between presuppositions and implicatures, we adopt the cover term 'inference' and will only make finer distinctions among the two inferences when necessary.

- (1) *Mary is the employee of the month, but we learned that the department where Mary works has been downsized due to a recent financial crisis:*
 Even [Mary]_F is fired.
 a. Assertion: Mary is fired.
 b. Additive inference: There is (at least) someone besides Mary who is fired.
 c. Scalar inference: Among the employees in the department, Mary is the most unlikely person that would be fired.

The two inferences, however, are not equally available in every *even*-sentence. For example, while the scalar inference in (2) is clearly available (winning the gold medal is more unlikely than winning other medals), there can hardly be an additive inference because the sentence does not infer that John has won any other medal (Rullman 1997; Lahiri 2008):

- (2) John even won [the gold medal]_F.

More examples will be discussed in Section 2.1, where it will be shown that either the scalar inference or the additive inference (but not both) can be suppressed.

Interestingly enough, it is observed that Mandarin Chinese has two lexical entries that correspond to *even* in English: *shenzhi* and *lian.dou* (consisting of *lian* ‘(lit.) include’ and *dou* ‘(lit.) all/every’) (see Shyu 2004, 2018 and [concealed authors] 2015):

- (3) a. Zhangsan shenzhi xihuan [Lisi]_F.
 Zhangsan even like Lisi
 ‘Zhangsan even likes Lisi.’ ≈ ‘It is even the case that Zhangsan likes Lisi.’
 b. Zhangsan lian [Lisi]_F dou xihuan.
 Zhangsan include Lisi all like
 ‘Zhangsan even likes Lisi.’ ≈ ‘Zhangsan likes even Lisi.’

We will treat the sentential adverbial *shenzhi* as a clausal EVEN because it can associate with any element inside the clause (the whole clause, the subject, the verb, the VP, or the object), and we refer to the particle complex *lian.dou* as a phrasal EVEN because the focused element is generally restricted to the local phrase sandwiched by *lian* and *dou* (usually an NP, but it can also be the predicate associated with *lian.dou*, as in Constant and Gu 2010).² Developing and sharpening the accounts in Shyu (2004, 2018) and [concealed authors] (2015), we argue that the clausal EVEN carries the scalar meaning inherently, and the additive inference is a

² As observed in Constant and Gu (2010: 28), the sandwiched phrase is not restricted to DP/NP:

- (i) a. Ta lian [chi]_F dou bu chi.
 3SG even eat all NEG eat
 ‘She does not even [eat]_F.’
 b. Ta lian [(jian) fumu]_F dou bu jian.
 3SG even see parent all NEG see
 ‘She does not even [see her parents]_F’ (...not to mention having dinner with them.)

In these cases, focus association is restricted to the locally modified VP sandwiched by *lian* and *dou*. (ib) indicates that sometimes the copied verb inside *lian.dou* can be omitted on the surface structure, but the focus still falls on the VP, as indicated by the alternatives that can be constructed from the sentence.

pragmatic implicature that can be canceled. On the other hand, the phrasal EVEN displays an opposite pattern. The phrasal EVEN is inherently additive, and it may carry the scalar inference by associating with a higher scalar operator. This distinction allows the phrasal EVEN (*lian.dou*) to occur in some contexts where its scalar inference seems trivialized. By examining the two EVENS in Mandarin Chinese, the goal of this paper is to tackle two of the lingering puzzles in the literature concerning the inferences of *even*: (i) Why can we cancel one of the inferences in certain contexts, and (ii) what is the nature of scale involved in the scalar inference?³

Our paper is organized as follows. First, we investigate the distinction between *shenzhi* and *lian.dou* in Section 2. We overview and summarize the problems of *even* in the literature and then apply the results to the two EVENS in Mandarin Chinese. Our analyses of the clausal EVEN *shenzhi* and the phrasal EVEN *lian.dou* are presented in Section 3. Section 4 reviews two recent accounts that attempt to derive scalarity from distributivity, or vice versa. We demonstrate that these accounts are inadequate and that our analysis of *lian.dou* provides a more straightforward alternative. In Section 5, we discuss how our analyses can have broader application to other languages that also employ different scalar additive particles to convey the EVEN meanings. Section 6 concludes the paper.

2. The EVEN duo in Mandarin Chinese: *shenzhi* and *lian.dou*

2.1 Setting the stage

To formalize the inferences triggered by *even*, we adopt the framework developed in Karttunen and Peters (1979), Rooth (1985, 1992), and Krifka (1991, 1998a), where *even* is a focus-sensitive particle that associates with a focused element in its scope, and the focus association triggers an alternative set contextually compatible with the focused element. Let us go back to (1), repeated below as (4):

- (4) *Mary is the employee of the month, but we learned that the department where Mary worked has been downsized due to a recent financial crisis:*
 Even [Mary]_F is fired.
 a. Assertion: Mary is fired.
 b. Additive inference: There is (at least) someone besides Mary who is fired.
 c. Scalar inference: For every employee in the department, Mary is the most unlikely person that would be fired.

Assume that Mary worked in a small department with only four employees – Mary, Sally, Tom, and John. The focus-induced alternative meaning will give rise to a set of alternative propositions in $C = \{\text{Mary is fired, Sally is fired, Tom is fired, John is fired}\}$. The additive inference then suggests that at least one alternative proposition besides the asserted one is true, and the scalar inference that it is most unlikely/unexpected that Mary is fired.⁴ The simplified formal representations are given in (5), where *even* is analyzed as a sentential

³ In Shyu (2018), the lack of additive/existential inference of *shenzhi* is due to the fact that *shenzhi* associates with the entire clause, while *lian.dou* only associates with the *lian*-NP (hence generating NP-alternatives that give rise to the additive inference) (Shyu 2018: 177). Shyu’s account, however, does not seem able to account for why the additive inference can still be missing in (3) and (16b), in which the NP is focus-marked, generating NP-alternatives.

⁴ We will see below that the scalar inference will be expanded to include a contextually supplied scale, as in Greenberg (2016, 2018).

operator, p is the prejacent proposition, and C is a contextually salient subset of p 's focus alternatives (see Rooth 1992):

- (5) *Even*(p)
- a. Assertion: p is true.
 - b. Additive inference: $\exists q \in C [q \neq p \ \& \ q \text{ is true}]$
(There is a q comparable but not identical to p , and q is true.)
 - c. Scalar inference: $\forall q \in C [q \neq p \rightarrow p <_{\text{likely}} q]$
(For each q comparable but not identical to p , p is more unlikely than q .)

For our purposes, we will focus on how the two inferences are triggered, the contexts or environments in which they appear, and the nature of the scale involved in the scalar inference. Turning back to the two inferences, both inferences are claimed to be available at the same time, but the claim is often challenged in the literature (Kay 1990; Lahiri 1998, 2008; Crnič 2011; Wagner 2015; Iatridou and Tatevosov 2016, among others).

Regarding the additive inference, it is observed that *even* may sometimes co-occur with *only* (Krifka 1991; Wilkinson 1996; Rullmann 1997; Crnič 2011; Wagner 2015). The following example, kindly provided by an editor, illustrates this phenomenon:⁵

- (6) John is a modest person. I expect him to drink only [beer]_F at the party, but in fact, **he even only drank [water]_F**.

According to Horn (1969), *only* carries an exclusive meaning, and it asserts that no other alternative propositions are true, i.e. $\neg \exists q \in C [q \neq p \ \& \ q \text{ is true}]$. The exclusive meaning of *only* is in effect the opposite of the additive meaning of *even*. However, (6) shows that *even* and *only* can co-occur in a sentence while sharing the same focus (see also Krifka 1991 and Wold 1996). This may suggest that it is the scalar inference of *even* that is triggered in these sentences, but not the additive one. Otherwise, the sentence should run into a contradiction (#John drank nothing but water, and he drank something other than water). Without the additive inference, (6) simply presupposes that it is most unlikely that John drank only water at the party (or we are extremely impressed by his level of modesty). The same problem is observed in (7) and (8) (from Lahiri 1998 and Szabolcsi 2017, respectively). If *even* triggered an additive inference, it would presuppose that John won more than one medal in a single competition (in addition to the gold medal), and someone else besides Eeyore also spat the thistles out (but in fact, Eeyore is the only one that ate them):

⁵ Francis (2019: 73) argues that the additive inference in (i), which is the original example discussed in Krifka (1991) and Wold (1996), remains available because *even* is associated with both the subject (through contrastive focus/topic) and the object, while *only* is associated with the object, as shown in (i). In this case, *only* will not produce mutually exclusive alternatives so it can co-occur with *even*:

(i) [John]_{F1} even₁ only₂ drank [water]_{F2, F1}.

The example in (6), however, is immune from this problem because it is unlikely for the subject to receive a contrastive interpretation in that context. Note that this type of sentence, which involves more than one focus-sensitive particle, poses a challenge to Rooth's alternative semantic approach because the lower focus operator would unselectively bind the variable generated by the focus-marked element, hence leaving no other variable for the higher focus operator to bind. This problem can be solved either by adopting a selective binding approach (Wold 1996) or by embracing a LF movement approach in conjunction with the alternative semantics (Rooth 1996, Krifka 2006). We will leave this complication aside in this paper, however.

- (7) John even won [the gold medal]_F.
 (8) *Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known for favoring thistles) takes a bite but spits it out.*
 Those thistles must be really prickly! Even Eeyore spat them out!

The problem is not restricted to the additive inference. In some cases, it is the scalar inference that seems to disappear, and only the additive inference is obtained. For example, the scalar inference seems to be canceled in the following sentences (Kay 1990, citing Fauconnier 1976):

- (9) a. George drank a little wine, a little brandy, a little rum, a little calvados, and even a little Armagnac.
 b. George drank a little wine, a little rum, a little Armagnac, a little calvados, and even a little brandy.

The sentences in (9), according to Kay (1990), do not concern any likelihood (or expectation) among the alcoholic drinks because the two sentences basically have the same truth condition and pragmatic effect. On the other hand, *even* here is close to the additive element like *plus* (Kay 1990: 74), which triggers only the additive inference. Kay (1990) argues that the scalar inference in this case comes from the additive inference because the final conjunct introduced by *even* not only increases the quantity or diversity of drinks but also serves a scale-closing function, thereby making the asserted sentence the most ‘informative’ (consuming five types of drinks entails consuming four types of drinks). This type of scalar inference is more trivial, however, because it is based on an almost tautological entailment (Adding some *x* to the conjoined sentence always entails the sentence without *x* no matter what *x* is). The question here is why this type of scalar inference does not involve a ‘subjective’ scale so that the last conjunct modified by *even* contributes to the most unlikely or noteworthy reading. It turns out that Mandarin Chinese may provide a clue to this question because different EVENs are employed (see the next subsection).

Another problem related to the scalar inference concerns the nature of the scale (whether it is simply based on likelihood). Greenberg (2016, 2018) offers considerable challenges to the likelihood-based analysis. Consider the following examples (from Greenberg 2016, 2018):

- (10) *A, an American businessman, has bought a new factory near Jerusalem.*
 A: I need someone in Israel to take care of the new factory.
 B: John lives in Israel. He even lives in [Jerusalem]_{F/#}[Ako]_F. (Ako is in the north of Israel.)
- (11) *Any princess who gives birth can stay in the palace. If she gives birth to a boy, she becomes a queen (with the result that on average 50% of those who have given birth become queens).*
 Princess Jane gave birth. She even gave birth to [a boy]_{F/#}[a girl]_F.
- (12) (*Seller to client:*) Both tools are strong. The one on the right is made of strong aluminum, and the one on the left is even made of [steel]_F.

Hiring someone from Jerusalem (a big city) can be more likely/expected than hiring someone from Ako (a small city), yet (10) is felicitous with *even* associating with *Jerusalem*. In (11), there is an equal chance of giving birth to a girl or a boy; nevertheless, *even* can be felicitously used for *a boy*, but not for *a girl*. The scenario in (12) poses a similar problem for the likelihood-based semantics of *even* because steel can be a more common material used to build tools than strong aluminum. Given these examples, Greenberg (2018) proposes a gradability-based account to replace the traditional likelihood-based account in (13), which simply compares the likelihood among the (alternative) propositions. Instead, she suggests that the scalar inference of *even* in English always involves a contextually supplied gradable property G (related to a non-focused part of the sentence). The sentence with *even* is salient when the maximal degree (d1) associated with the (accessible) worlds in which the prejacent proposition (p) is true exceeds the maximal degree (d2) associated with every (accessible) world in which the alternative proposition (q) is true and p is false (the exhaustified q-worlds). (13) and (14) illustrate the likelihood-based and the gradability-based scalar inferences, respectively:

- (13) Likelihood-based scalar inference: p is more unlikely than q

----- q ----<----p ----- $\rightarrow$ unlikely

- (14) Gradability-based scalar inference: For a salient entity x (denoted by some non-focused constituent in p), x's degree on a G scale in the (accessible) p worlds is higher than x's degree in the accessible q-and-not-p worlds (where both d1 and d2 are at least as high as the standard of G):⁶

$$\begin{array}{ccc} & w2: q \text{ and } \neg p & w1: p \\ & \uparrow & \uparrow \\ \text{----- standard -----} & x\text{'s degree (d2)} & \text{---- } x\text{'s degree (d1)} \text{ ----} \end{array} \rightarrow G$$

Consider the case in (10). In this scenario, a suitable G can be the 'degree of convenience for John to go to work'. In w1 (p=John lives in Jerusalem), John's degree of convenience exceeds his degree of convenience in the alternative worlds w2 (q=John lives in Ako and does not live in Jerusalem). Therefore, the use of *even* is felicitous here. Cases like (11) and (12) are similar. The relevant G in (11) can be 'Jane's degree of happiness (or importance in the royal house)', and the speaker is comparing the worlds between w1 (p=Jane gives birth to a boy) and w2 (q=Jane does not give birth to a boy, and Jane gives birth to a girl or does not have any child). The relevant G in (12) can be about 'the strength of the tools.'

Zhang (2022) provides further evidence for Greenberg's proposal. Zhang (2022) notices that *even*-sentences in English always address a contextually salient degree-related Question Under Discussion (QUD) about the non-focused part of the *even*-sentences. On the other hand, *only/also*-sentences address a QUD about the focused part. Consider the following contrasts under the scenario given by Szabolcsi (2017):⁷

⁶ For expository convenience, we will neglect the requirement that x's degrees in the p-worlds and that in the q-but-not-p-worlds should exceed the standard degree of G in the following discussion.

⁷ A reviewer has noted that the subject of (15b) should be understood as a contrastive topic, which is analyzed as a topic phrase that contains a focus phrase inside (Krifka 1998b).

- (15) *Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known to favor thistles) takes a bite but spits it out:*
- Only [Eeyore]_F spat thistles out (and no one else did).
(QUD: Who spat out thistles?)
 - (Pooh spat thistles out.) [Eeyore]_F also spat thistles out.
(QUD: Who spat out thistles?)
 - (Those thistles must be really prickly!) Even [Eeyore]_F spat thistles out.
(QUD: How prickly are those thistles?)

We will return to this point later, where we observe that most of the examples mentioned in Greenberg (2016, 2018) are best compatible with the clausal EVEN *shenzhi* in Mandarin Chinese, but much less so with the phrasal EVEN *lian.dou*. This suggests that the scales involved in the scalar inference are determined by the types of EVEN used in the sentences, and this helps settle the issue on the nature of the scale in the scalar inference.

Finally, it might be suggested that both of the so-called inferences are actually pragmatic implicatures in the sense of Grice (1975), so they might be cancelable/suspended in different contexts. However, although one of the two inferences can be suspended in one case or another, the other is obligatorily triggered. This indicates that they are not simply pragmatic implicatures; otherwise, we should be able to find a context where both implicatures were canceled (but in fact we cannot).

2.2 Severing the two inferences in Mandarin Chinese

Recall that our assumption is that *shenzhi* always carries a scalar inference (but not necessarily the additive one) and *lian.dou* always carries an additive inference (but the scalar one can be less obvious). The assumption can be verified by constructing various pragmatic contexts. There are contexts in which the two inferences are mutually exclusive/incompatible (Rullman 1997; Crnič 2011), so the additive inference may lead to a contradiction, and we will see that only *shenzhi*, but not *lian.dou*, is felicitous in these contexts. At the same time, there are also contexts in which the scalar inference of *even* becomes more obscure, and as predicted, these contexts reject the use of the scalar *shenzhi* but welcome the additive *lian.dou*.

The first case that excludes the use of the additive *lian.dou* is when there is an exclusive particle *zhi*(-you) ‘only’:

- (16) Co-occurrence with the exclusive *only*
- Zhangsan is a modest person, and I expect him to only drink beer...*
- Shishishang, *ta lian zhi-you [shui]_F dou he.
In.fact 3SG even only-have water all drink
Intended: ‘In fact, he even drank only [water]_F.’
 - Shishishang, (ta) shenzhi zhi he [shui]_F.
In.fact 3SG even only drink water
‘In fact, he even drank only [water]_F.’

The exclusive particle *zhi* ‘only’ in Mandarin Chinese cannot directly modify an NP (**zhi shui* ‘only water’). Therefore, a spurious particle *you* ‘(lit.) have’ is needed in (16a). As shown in (16a), the additive particle *lian.dou* and the exclusive particle *zhi* cannot co-occur while sharing the same focus (*water* in this case) because such a configuration leads to a

semantic clash. The scalar particle *shenzhi* in (16b), on the other hand, is acceptable with the exclusive particle because *shenzhi* does not trigger any additive inference that may result in a contradiction. Notice that (16b), in which *shenzhi* ‘even’ and *zhi* ‘only’ share the same focus on ‘water’, might not represent the most natural interpretation of the sentence. Different focus association patterns are possible under different contexts (see Krifka 1991 for discussion), but we choose the context in (16) to force *shenzhi* ‘even’ and *zhi* ‘only’ to share the same focus in order to highlight the fact that the additive inference of *shenzhi* can be canceled under certain contexts.⁸ (16b) also allows the subject to be *pro*-dropped, and this lends further support to the fact that the subject focus is not involved in this type of sentence (see footnote 5; we thank an editor for pointing this out to us).

The second case also involves an exclusive context (our world knowledge suggests that there is only one person who may win first place in an exam). Again, the sentence is acceptable with the scalar particle *shenzhi* (17a) (winning first place is most noteworthy considering John’s daughter’s achievements), and it is not acceptable with the additive particle *lian.dou* (17b):⁹

(17) The exclusive context

John’s daughter performed exceptionally well in the exam last week:

a. Ta shenzhi dedao-le [di-yi-ming]_F.

3SG even obtain-PFV the-first-place

‘It is even the case that she won first place.’

b. ??Ta lian [di-yi-ming]_F dou dedao-le.

3SG even the-first-place all obtain-PFV

‘She even won first place.’

(Okay if the context is changed to that she won second place in the last exam and first place in this exam, which has an additive inference.)

The mutual exclusiveness among alternatives also rules out the use of the additive EVEN *lian.dou* when it associates with a universal QP (18b) (see also Wagner 2015), but the scalar

⁸ A reviewer asks whether (16a) can be ruled out on syntactic grounds (for example, *lian* must occur immediately before its focus). Such concern can be dispelled in (i). When the particle *zhi...* (*eryi*) has a scalar meaning ‘just/as little as’ (the sentence does not infer that there is only one bowl of rice, but the quantity of rice is minimal), it can co-occur with *lian.dou*:

(i) Zhangsan lian [zhi-chi yi-wan_F (eryi)] dou chi-bu-wan.
Zhangsan include just-eat one-CLF only all eat-NEG-finish
‘Zhangsan cannot finish even just one bowl (of rice).’

The example suggests that (16a) is ruled out by semantic clash, rather than syntactic ill-formedness. Furthermore, the reviewer suggests that it is more natural for *shenzhi* to associate with the VP in (16b), but we believe that (16b) is valid within the presented context, as it can be paraphrased into the *but* construction in (ii), which suggests that it is *shui* ‘water’ that bears focus (Drubig 2003). When the VP bears focus, it would be paraphrased into (iii), which does not convey the intended meaning given the context in (16):

(ii) Zhangsan shenzhi shenme dou mei-he, zhi he [shui]_F.
Zhangsan even what all NEG-drink only drink water
‘Zhangsan even drank nothing but (only) water_F.’

(iii) Zhangsan shenzhi shenme dou mei-zuo, zhi [he shui]_F.
Zhangsan even what all NEG-do only drink water
‘Zhangsan even did nothing, but only [drank water]_F.’

⁹ Judgments of the examples in Mandarin Chinese (unless otherwise cited) are based on our own native intuitions. We also consulted three other native speakers of Mandarin Chinese (spoken in Taiwan), and all of them agree with our judgments reported in this paper.

EVEN *shenzhi* can be used in this case (18a). (18c) shows that additivity does not go along with a universal QP, so the additive particle *ye* ‘also’ cannot associate with *suoyou* ‘all’. If ‘also’ indeed occurs with a universal QP, the focus association must shift to the verb (18d) (the contrast falls on the verb instead – there are other things that the teacher did to all students than failing them) or other constituents than the universal QP:

(18) Association with a universal quantifier

The qualities of the students are getting worse, and this year, students are so awful that...

- a. Laoshi shenzhi gei-le [suoyou]_F de xuesheng bu-jige.
 teacher even give-PFV all DE student NEG-pass
 (bu-shi cai gei-le yi-ban de xuesheng.)
 NEG-be just give-PFV half DE student
 ‘It is even the case that the teacher failed all of the students (not just half of them).’
- b. #Laoshi lian [suoyou]_F de xuesheng dou gei-le bu-jige.
 teacher even all DE student all give-PFV NEG-pass
 ‘#The teacher failed even [all]_F of the students.’ (implies ‘The teacher also failed most/some of the students’)
- c. # (Bu-shi cai yi-ban de xuesheng;) laoshi ye gei-le
 NEG-be just half DE student teacher also give-PFV
 [suoyou]_F de xuesheng bu-jige.
 all DE student NEG-pass
 Intended: ‘(Not just half of them) the teacher also failed all of the students.’
- d. (Bu-zhi ma-le tamen;) laoshi ye [gei-le]_F suoyou
 NEG-only scold-PFV 3PL teacher also give-PFV all
 de xuesheng bu-jige.
 DE student NEG-pass
 ‘(Not only did the teacher scold them) the teacher also [failed]_F all of the students.’

Lastly, there are the types of sentences (10)–(12) originally brought up by Greenberg (2018). Their Mandarin Chinese counterparts are listed in (19)–(21), respectively. Interestingly, each of the sentences is compatible with *shenzhi*, but much less so with *lian.dou*:

(19) *A, an American businessman, has opened a new office in Taipei, and he needs someone in Taiwan to run the business there.*

- a. Zhangsan zhu-zai Taiwan. Ta shenzhi zhu-zai [Taipei]_F.
 Zhangsan live-in Taiwan 3SG even live-in Taipei
 ‘Zhangsan lives in Taiwan. He even lives in Taipei.’
- b. #Zhangsan zhu-zai Taiwan. Ta lian [Taipei]_F dou zhu.
 Zhangsan live-in Taiwan 3SG include Taipei all live
 Intended: ‘Zhangsan lives in Taiwan. He even lives in Taipei.’¹⁰

¹⁰ The sentence with *lian.dou* is acceptable when one is talking about how many places Zhangsan has lived in:

(i) (Zhangsan zhu-guo henduo difang.) Ta lian Taipei dou zhu-guo.
 Zhangsan live-EXP many place 3SG include Taipei all live-EXP
 ‘(Zhangsan has lived in many places.) He even lived in Taipei before.’

Another way of expressing this sentence with *lian.dou* is like (ii), but this suggests that there are other alternatives (such as Zhangsan’s work or school) that are also in Taipei:

(20) *Princess Jane just gave birth to a child...*

- a. Ta shenzhi sheng-le [nanhai]_F.
 3SG even give.birth-PFV boy
 ‘It is even the case that she gave birth to a boy.’
- b. #Ta lian [nanhai]_F dou sheng-le.
 3SG even boy all give.birth-PFV
 ‘She gave birth even to a boy.’¹¹

(21) (*Seller to client:*) *Both tools are strong. The one on the right is made of strong aluminum, and...*

- a. Zuo-bian-de shenzhi shi yong [gang]_F zuo de.
 left-side-DE even be use steel make DE
 ‘The one on the left is even made of [steel]_F.’
- b. #Zuo-bian-de lian [gang]_F dou yong-le.¹²
 left-side-DE even steel all use-PFV
 (The sentence is fine if the tool is made of **both** strong aluminum and steel.)

Note that these contexts are not compatible with *lian.dou* because none of them involve an additive inference (living in Taipei does not suggest living in another city in Taiwan, giving birth to a boy does not imply giving birth to another child, and being made of steel is a unique property of the tool).

Let us turn to the opposite situation. We mentioned the listing sentences in (9) (Fauconnier 1976; Kay 1990), where the last conjunct introduced by *even* seems to carry only the additive inference, and the scalar inference either goes missing or involves a more trivial

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- (ii) Zhangsan lian [jia]_F dou zai Taipei.
 Zhangsan include home all in Taipei
 ‘Even Zhangsan’s home is in Taipei.’

¹¹ Chen and Greenberg (2022) also note the similarity between *shenzhi* in Mandarin Chinese and *even* in English. Meanwhile, they use (i) to show that *lian.dou* can also be used in a context like (20). For them, *lian.dou* and *shenzhi* seem to be interchangeable:

- (i) *A, unaware of whether Princess Jane has become a queen or not, asks B whether she has children, and B answers:*

B: Ta you, (lian) [erzi]_F dou you.
 3SG have include son DOU have
 ‘She has (children), and she even has a son.’

The context in (i), however, is different from the original context, and for us, the sentence most naturally presupposes that Princess Jane has more than one child (and at least one of them is a boy), so the additive inference stays in this case (the original context in (20) excludes this reading because it is about the child Princess Jane just gave birth to). Since Chen and Greenberg (2022) do not discuss other related examples in their paper, we do not know how robust their judgments are regarding the similarities between *shenzhi* and *lian.dou*.

¹² An anonymous reviewer points out that (i) below is acceptable with *dou* ‘all’ (but not *ye* ‘also’). However, we suspect that there is a dialectal distinction here because (i) is generally considered unacceptable in our own variety of Mandarin Chinese:

- (i) %Zuo-bian-de dou yong-le gang.
 left-side-DE even/all use-PFV steel
 ‘The one on the left is even made of steel.’

We use *hai* ‘still’ instead of *dou* ‘all’ in our variety. Interestingly, this suggests that *dou* (but not *ye*) may have acquired other scalar uses in different dialects or varieties of (Mandarin) Chinese. We will leave this topic for future research.

scalar relation (based on the increased quantity of information, regardless of the likelihood of the item associated with *even*). In Mandarin Chinese, the corresponding sentence is produced in (22), in which the additive *lian.dou* can be used (confirming Kay’s observation that this type of *even* is akin to the additive *plus*), as in (22a), and the meanings do not change much whether the last conjunct is ‘beer’ or ‘rice wine’ (both are considered common types of alcoholic drinks in Asia). On the other hand, if *shenzhi* is used in the last conjunct, as in (22b), then the worlds in which the prejacent proposition ‘Zhangsan drank beer’ holds must exceed the alternative worlds (where Zhangsan drank other types of alcoholic drinks but did not drink beer) with respect to a contextually-supplied gradable property G, and the suggested context in (22) does not suit this requirement. (22b) shows that *shenzhi* can only be used in a context where it is atypical of Zhangsan to drink beer, and drinking beer is considered noteworthy with respect to some G (e.g., Zhangsan’s desperateness):

(22) Listing sentences: *Zhangsan drank so many different types of alcoholic drinks at the party last night. Let me tell you what he drank...*

- a. Zhangsan bu-dan he-le xiangbin, hong-jiu, bai-jiu, he
 Zhangsan NEG-only drink-PFV champagne red-wine white-wine and
 pijiu/mijiu, lian mijiu/pijiu dou he-le bushao.
 beer/rice.wine include rice.wine/beer all drink-PFV some
 ‘Zhangsan drank champagne, red wine, white wine, beer/rice wine, and even some rice wine/beer.’
- b. ^{??}Zhangsan bu-dan he-le xiangbin, hong-jiu, bai-jiu, he
 Zhangsan NEG-only drink-PFV champagne red-wine white-wine and
 mijiu, shenzhi he-le bushao pijiu.
 rice.wine even drink-PFV some beer
 ‘Zhangsan drank champagne, red wine, white wine, rice wine, and it is even the case that he drank some beer.’
 (Okay if Zhangsan does not usually drink beer, and he is so desperately in need of alcohol that he even drank beer.)

The overall results confirm our initial hypothesis – *shenzhi* always carries a scalar inference (based on a contextually supplied gradable property that addresses a non-focused part of the sentence), and its additive inference can be canceled. What is more, *lian.dou* displays an almost opposite pattern – the additive inference is always there while the scalar inference is sometimes more trivial.

It should also be pointed out that in contexts where the two inferences do not result in contradiction or infelicities, *shenzhi* and *lian.dou* can co-occur in a sentence, which is commonly found in Mandarin Chinese. In such cases, the two inferences are both available:

- (23) Lisi **shenzhi** **lian** liulian **dou** chi.
 Lisi even include durian all eat
 ‘Lisi even eats durians.’ (with both additive and scalar inferences)

This is not a problem for our generalization, though. Our point is that *lian.dou* always calls for the additive reading, while the scalar inference can also be observed in appropriate contexts, and in contexts that generate both additive and scalar inferences, *shenzhi* and *lian.dou* can co-occur in a sentence.

2.3 The syntactic distinctions

In addition to the semantic/pragmatic differences, we observe that the clausal EVEN *shenzhi* is banned from certain syntactic environments, while these environments generally allow the use of the phrasal EVEN *lian.dou*. Specifically, *shenzhi* cannot occur in imperatives or in antecedents of conditional sentences. These conditions suggest that the clausal *shenzhi* is an instance of the main clause phenomena (Haegeman 2012) (we thank Manfred Krifka p.c. for pointing this out to us):

(24) Imperative sentences:

- a. *Shenzhi chi-le shengxia de huasheng ba!
even eat-PFV rest DE peanut SFP
- b. Lian shengxia de huasheng dou chi-le ba!
include rest DE peanut all eat-PFVSFP
'Eat the rest of the peanuts, too (will you)!'

(25) Antecedents of conditional clauses:

- a. ^{??}Ruguo Lisi shenzhi likai-le, wo hui geng gaoxing.
if Lisi even leave-PFV 1SG will more happy
- b. Ruguo lian Lisi dou likai-le, wo hui geng gaoxing.
if include Lisi all leave-PFV 1SG will more happy
'If Lisi also left, I would be much happier.'

In the next two sections, we develop a syntax-semantics account to explain the different syntactic distributions of the two EVENS in Mandarin Chinese and why they can give rise to different types of inferences.

3. Analyses of the two EVENS in Mandarin Chinese

3.1 *Shenzhi* as a speaker-oriented element

Shyu (2004, 2018) proposes that *shenzhi* patterns with speaker-oriented adverbs that express the speaker's unexpectedness, such as *jingran/juran* 'unexpectedly.' Following this lead, we assume that *shenzhi* is analyzed as a speaker-oriented particle (although *shenzhi* operates on a more general scale that is not limited to the speaker's own expectation; see examples above). Treating *shenzhi* as a speaker-oriented element has the advantage of accounting for its syntactic distributions. Recall that *shenzhi* shows strong selection on the sentential mood (24)–(25). These properties are cross-linguistically shared by other speaker-oriented elements, such as *fortunately* and *luckily* in English, or *xingkui* 'luckily' in Mandarin Chinese, as shown in (26)–(27) (see Bellert 1977; Ernst 2002, 2009; Haegeman 2012; Liu 2012; Mayol and Castroviejo 2013). Note that it has been pointed out in Haegeman (2012) and Liu (2012) that factivity plays a crucial role in determining whether a speaker-oriented adverb may appear in the conditional antecedent. For our purpose here, let us consider the central conditional clause with the speaker-oriented adverb *xingkui* 'luckily', which, like *shenzhi*, implies the truth of its prejacent proposition (see Liu 2012 for further discussion). As predicted, such combinations lead to ungrammaticality:

- (26) a. *If John luckily arrives, he may stay in this room.
- b. *Please luckily arrive on time.

- (27) a. *Ruguo Lisi xingkui/shenzhi lai-le,
if Lisi luckily/even arrive-PFV
ta keyi zhu-zai zhe ge fangjian.

3SG can stay-in this CLF room
 Intended: ‘If Lisi luckily/even arrives, he may stay in this room.’

- b. *Qing xingkui/shenzhi zhunshi lai.
 please luckily/even on.time arrive
 Intended: ‘Please luckily/even arrive on time!’

Besides the syntactic distribution, our analysis also accounts for why *shenzhi* always calls for a scalar inference that evaluates accessible possible worlds with respect to a contextually salient G. In terms of their meanings, speaker-oriented adverbs, such as *guoran* ‘expectedly’ and *jingran/juran* ‘unexpectedly’, do not change the truth conditions of the modified sentences, but they carry inferences concerning the speaker’s expectation. Specifically, the meaning of these adverbs can be decomposed into two parts: the first part contains a doxastic attitude of the speaker (the speaker expects/believes that...), and the second part concerns the truth of the proposition in the evaluation world (Ernst 2009). As a result, these adverbs express the (non-)congruency of the modified proposition in the doxastically accessible worlds (w’) and in the actual world (w). Adopting the possible world semantics of attitude predicates (Thomason and Stalnaker 1973; Kratzer 1981; Heim and Kratzer 1998; Winter 2016), we propose that sentences with a mirative adverb, like *juran* or *jingran* ‘unexpectedly’, express a contrast between the assertion and the inference, as in (28).

- (28) Zhangsan juran/jingran xihuan Lisi.
 Zhangsan unexpectedly like Lisi
 ‘Unexpectedly, Zhangsan likes Lisi.’
 a. Assertion: Zhangsan likes Lisi in w.
 b. Inference: $\forall w'$ such that w' is accessible to w (w' is compatible with the speaker's expectation in w), it is not the case that Zhangsan likes Lisi in w'.

The speaker-oriented *shenzhi* can be viewed as a type of speaker-oriented particle that involves an additional layer of complication due to its comparative meaning (Greenberg 2018). That is, the speaker compares different accessible worlds when they are mapped to a contextual gradable property G. To illustrate, let us take (29) as a concrete example. Given the focus-induced alternative semantics (Rooth 1992), a set of alternative propositions is generated by replacing the focus-marked Lisi with each of the contextually relevant persons. The result is the set of alternative propositions in (29b). Applying the gradability-based semantics (Greenberg 2018), we see that the clausal EVEN *shenzhi* introduces a comparative meaning among the (accessible) possible worlds with respect to G. The relevant G in (29) can be about ‘Zhangsan’s tolerance in treating people’. Eventually, (29c) means that Zhangsan is considered more tolerant in w1 (in which he likes Lisi) than in w2 (in which he likes Wangwu and/or Mary but he does not like Lisi):

- (29) *We are talking about Zhangsan’s good personality and his relationship with his colleagues, Lisi, Wangwu, and Mary. Meanwhile, Lisi is a snob who few people like. Zhangsan is such a tolerant person that...*
 Zhangsan shenzhi xihuan [Lisi]_F.
 Zhangsan even like Lisi
 ‘Zhangsan even likes Lisi.’
 a. Assertion: Zhangsan likes Lisi in w.

- b. Alternative propositions (generated by replacing [Lisi]_F with contextually relevant entities): {Zhangsan likes Lisi, Zhangsan likes Wangwu, Zhangsan likes Mary}
- c. Scalar Inference: $\forall w_1 \forall w_2$ [[w_1 and w_2 are accessible to w , and Zhangsan likes Lisi in w_1 , and Zhangsan likes Wangwu and/or Mary, but not Lisi in w_2] $\rightarrow$ [The maximal degree regarding G in w_1 exceeds that in w_2 .]]

Notice that in w_1 , it is not defined whether Zhangsan likes Wangwu and/or Mary, but since the accessibility condition is based on what is considered normal/typical by the speaker in the actual world w , it can be implied that if Zhangsan likes Lisi, he also likes Wangwu and Mary (since Zhangsan is considered an exceptionally tolerant person) in any normal and typical worlds, and this may include the actual world. Therefore, the sentence may have an additive inference, but such an inference can be canceled when the alternatives are mutually incompatible (e.g., in the gold-medal context), in which case the alternative propositions and the prejacent propositions cannot both be true at the same time. This explains why the additive inference can be canceled when *shenzhi* is used.¹³

3.2 On *lian.dou*: From Additivity to Scalarity

Earlier analyses of *lian.dou* generally assume that either *lian* or *dou* is a scalar item that occupies the head of the scalar focus phrase and is responsible for the scalar meaning (see M. Xiang 2008; Badan and Del Gobbo 2015; Liu 2016; among others). The main idea of these analyses is that *lian.dou* triggers a focus movement of XP to the Scalar Phrase, which gives rise to the scalar inference:

- (30) The inherent scalar analysis of *lian.dou* (to be revised)
 Zhangsan [Focus/ScalarP *lian* [*dangao*] *dou* mei-chi [~~*dangao*~~]].
 Zhangsan include cake all NEG-eat
 ‘Zhangsan did not even eat the cake.’

The analysis, however, faces at least two problems. First, it does not account for why *lian.dou* and *shenzhi* may bring about different inferences (as we have seen earlier) if both correspond to the same *even* particle. Second, the focus-movement analysis generally does not consider the original lexical meanings of *dou* and *lian*, and it simply assumes that *dou* or *lian* (or both) may correspond to *even* in English. However, the mechanism behind the scalar inference remains obscure.

In this section, we develop an account that aims to address both questions. We show that both *lian* and *dou* contribute to the additive and scalar meanings of EVEN through the mechanism known as domain widening (Kadmon and Landman 1993; Zanuttini and Portner 2003; Shank 2004; Chierchia 2006, 2013, among others). The basic idea is that the universal/distributive particle *dou* (Lin 1998) introduces a resource domain (von Stechow 1994) that contains typical entities in a given context. This domain is then modified by the *lian*-phrase, which adds a new entity to the initial set. Finally, a widening operator defines the newly added entity as the most unlikely (or informative) one, thus resulting in an EVEN meaning through domain widening.

¹³ In this sense, the scalar inference associated with *shenzhi* can be analyzed as a presupposition or a conventional implicature (Potts 2005; Liu 2012), while its additive inference can be regarded as a conversational implicature that can be canceled in certain contexts.

The domain-widening effect (Kadmon and Landman 1993) was originally designed to capture a semantic-pragmatic effect associated with the strong polarity item *ANY* in English (the capital letter indicates an emphatic reading), which widens the resource domain associated with the weak polarity item *any*. For example, with an emphatic *ANY*, (31d) becomes a stronger proposition than (31b) after domain widening since the domain now may include wet or dirty socks:

- (31) a. A: Could I borrow some socks?
 b. B: I don't have any socks.
 c. A: I don't mind if they are slightly damp.
 d. B: I don't have ANY socks (including wet or dirty ones).

Later analyses argue that domain widening can be attributed to the focus association of *any* with an emphatic focus operator, which acts like a covert EVEN (Krifka 1995a).¹⁴ In this sense, (31d) has a reading similar to *It is even the case that I don't have any wet or dirty socks (not to mention clean socks)*. In addition to NPIs, domain widening has also been adopted to account for a wider range of phenomena (Lahiri 1998; Kratzer and Shimoyama 2002; Zanuttini and Portner 2003; Shank 2004, Chierchia 2013), including indeterminate pronouns, free choice items, and exclamation. For example, Zanuttini and Portner (2003) argue that an exclamative sentence like (32) contains a widening operator R_{widening} , which has the function of widening the initial domain (D1) to a larger domain (D2) that contains at least one entity (i.e., habanero) that is the most unlikely to be included in D1:¹⁵

- (32) *We are discussing which hot peppers some of our friends like to eat. Most of them eat poblanos, serranos, jalapeño, and güero, but one of these friends even eats habanero (an extremely spicy pepper that people do not normally eat):*
 a. Che roba che l magna! (Italian: from Zanuttini and Portner 2003: 50)
 what stuff that 3SG eats
 'The things he eats!'
 b. Initial domain (D1) = {poblanos, serranos, jalapeño, güero}
 c. Widened domain (D2) = {poblanos, serranos, jalapeño, güero, habanero}

Zanuttini and Portner (2003) formulate the widening effect in (33) below. (33i) ensures that widening adds some item(s) to the initial domain (D1), while (33ii) says that the newly added item(s) are considered more extreme on some relevant scale (the symbol $<$ represents an ordering on a subset of the domain):

¹⁴ Krifka (1995a) calls it an emphatic assertion (Emph.Assert) operator. Under the structured meaning approach, the meaning of the operator is shown in (i), where B stands for the background, F for the focused item, A for alternatives, and $c \cap B(F)$ is the function of a typical assertion (to update the context c with the given proposition B(F)):

(i) **Emph.Assert** ($\langle B, F, A \rangle$)(c) = $c \cap B(F)$, iff:

(a) For all $F' \in A$: $c \cap B(F) <_c c \cap B(F')$

(b) $c \cap B(F) <_c c \cap \{c \cap B(F') \mid F' \in A\}$

(where $X <_c Y$ means X is less likely/more informative than Y given the common ground c).

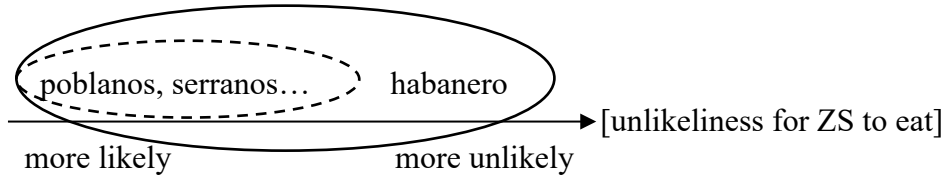
(i-a) says that the given proposition is less likely/more informative than any of the alternative propositions in the context, and (i-b) says that the given proposition is less likely than the conjunction of the alternative propositions. See Krifka (1995a) for details.

¹⁵ We are indebted to an anonymous reviewer for much of the discussion in this section, who urged us to clarify the function of the emphatic operator from the perspective of Zanuttini and Portner (2003).

- (33) WIDENING: For any clause S containing R_{widening} , widen the initial domain of quantification for R_{widening} , $D1$, to a new domain, $D2$, such that
- (i) $\llbracket S \rrbracket_{w, D2} < - \llbracket S \rrbracket_{w, D1} < \neq 0$ and
 - (ii) $\forall x \forall y [(x \in D1 \ \& \ y \in (D2 \setminus D1)) \rightarrow x < y]$.

Against this background, we argue that the *lian.dou* construction in Mandarin Chinese also contains a widening operator, which we represent as a covert Emphatic Assertion (Emph.Assert) operator, following Krifka (1995a), Lahiri (1998) and Hole (2004).¹⁶ Furthermore, since *lian* literally means ‘include’, the *lian*-phrase functions by adding an item to the initial resource domain, satisfying the condition in (33i), and the Emph.Assert operator then defines the newly added item as the most unlikely item to be considered, as in (33ii). The domain-widening analysis thus captures the intuition that the function of *lian* is to introduce an extreme item on the scale (Shyu 2016; Yang 2018). At the same time, it accounts for the semantic source of the scalar meaning. We illustrate the basic concept in (34):

- (34) a. Lajiao, [Emph.Assert Zhangsan [lian habanero]_F dou chi].
 pepper Zhangsan include habanero all eat
 ‘As for peppers, Zhangsan ate (any pepper) including habanero.’
 b. The domains of *lajiao* ‘pepper’:



- c. Initial domain ($D1$) = {poblanos, serranos, jalapeño, güero}
- d. Widened domain ($D2$) = {poblanos, serranos, jalapeño, güero, habanero}

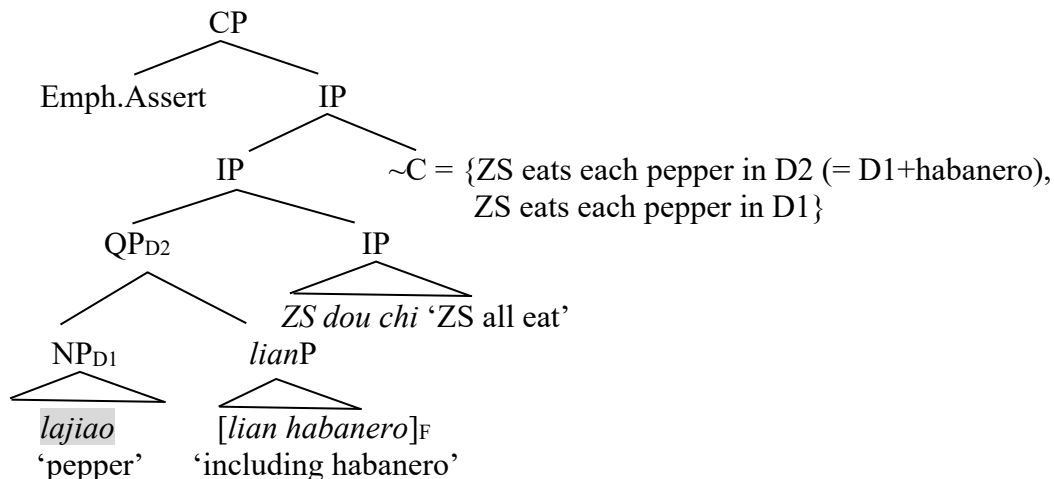
Technically, we follow Shank’s (2004) proposal that domain widening is triggered when a focus operator associates with the resource domain of a quantifier like *any* or *every* (so that when we say *Everyone is here*, it means that ‘every relevant person in this context is here’). In terms of notation, we use the subscript D to represent the resource domain restriction associated with a QP/NP. Adopting Rooth’s (1992) theory of alternative semantics, Shank argues that focus association generates alternatives that replace the resource domain of the quantifier with smaller domains in the contextual alternative set C (see also Chierchia 2013).¹⁷ We slightly modify Shank’s analysis to accommodate the *lian*-phrase in Mandarin Chinese. First, we assume that a covert bare NP-phrase (this could also be a *renhe*-NP ‘any NP’ or an indeterminate *wh*-phrase) is associated with *dou*, which provides the initial resource domain ($D1$). Second, the *lian*-phrase has the function of including new entities in $D1$ and thus increases the domain from $D1$ to $D2$. Finally, the Emph.Assert focus operator

¹⁶ Hole (2004: 110) specifically points out that the Emph.Assert operator can be employed to analyze the *lian.dou* construction in Mandarin Chinese. Our theory builds on this idea and offers a more formally explicit way of embodying this idea through the concept of domain widening.

¹⁷ For Shank (2004), the semantic-pragmatic function of domain widening is to cancel the scalar implicature of the smaller domain. Consider (34) again. By scalar implicature, the speaker may assume that Zhangsan only eats the peppers in the smaller domain $D1$. Domain widening, on the other hand, cancels the scalar implicature by adding a more unlikely entity (habanero) into the domain of *any*, hence creating a meaning akin to EVEN.

associates with the *lian*-phrase, and the focus association generates two relevant alternatives in C, a proposition with a larger resource domain D2 (including habanero) and another proposition targeting a smaller resource domain D1 (not including habanero). We illustrate the LF structure in (35a) (the shaded part represented the unpronounced QP/NP associated with *dou*). (35b) illustrates the meaning of the sentence with *dou*, which conveys a distributive or universal meaning, following the influential analysis of Lin (1998).¹⁸ (35c) is the denotation of the indefinite bare NP [*pepper*], which represents a set of peppers with a resource domain D1 (Krifka 1995b; Cheng & Sybesma 1999). (35d) spells out the function of *lian*, which creates a new domain D2 via adding an entity to the initial domain D1:¹⁹

(35) a. LF of *lian.dou* construction



- b. $\llbracket \text{Zhangsan dou chi} \rrbracket = \lambda X. \forall y[y \in X \rightarrow \text{ZS eats } y]$
(from Lin 1998: universal/distributive meaning of *dou*)
- c. $\llbracket \text{NP}_{D1} \text{ lajiao} \rrbracket = \lambda x. x \text{ is a pepper in } D1$
- d. $\llbracket \text{QP}_{D2} \rrbracket = \llbracket \text{NP}_{D1} + \text{lian-NP} \rrbracket = \lambda x. x \text{ is a pepper in } D2, \text{ including habanero}$
- e. $\llbracket [\text{IP Zhangsan lian habanero dou chi}] \rrbracket = \forall y[y \text{ is a pepper in } D2 \rightarrow \text{ZS eats } y]$
= {ZS eat poblanos, ZS eats serranos, ZS eats jalapeño, ZS eats güero, ZS eats habanero}

Finally, (35e) means that ‘Zhangsan eats each pepper in D2, including habanero’, and it correctly derives the assertive meaning of (34) (which is equivalent to the preajcent excluding the focus operator), which encodes the additive inference as part of the lexical meaning of *dou*. As for the scalar inference, it comes from focus association of the *lian*-phrase with the Emph.Assert operator that operates on the alternatives in C. Following Shank (2004: 33), we translate the Empha.Assert operator in Krifka (1995a) below (see also Eckardt 2001: 389):

¹⁸ While we assume that *dou* ‘all’ and *ye* ‘also’ (see Section 3.4 below) may keep their original universal/distributive and additive meanings in the *lian.dou/ye* construction, we are not committed to the claim that these parametric uses of *dou/ye* share the same syntactic distributions as the original ones (see Hole 2004 and Yang 2020 for discussion of this issue). We will leave this question pending for future research.

¹⁹ One could reformulate the NP_{D1} as a A-bar bound *pro* or a silent restriction that is bound by a (dangling) topic in the left periphery, as illustrated in (i) below. We thank an anonymous reviewer for pointing this out to us.

(i) Shuo-dao lajiao_i, Zhangsan [(sheneme) lajiao]_i dou chi.
speaking pepper Zhangsan what pepper all eat
‘As for peppers, Zhangsan eat each pepper.’

- (36) Scalar inference of (34): from Emph.Assert
 $\llbracket (36) \rrbracket$ is defined iff
 $\forall q[[q \in C \ \& \ q \neq \phi] \rightarrow \phi <_{\text{likely}} q]$
 (where ϕ = ‘Zhangsan eats each pepper in D2, including habanero.’)

Here, the asserted proposition ϕ [Zhangsan eats each pepper in D2] is considered more unlikely than the alternative proposition q [Zhangsan eats each pepper in D1] due to the inclusion of habanero, which is a pepper that is most unlikely to be consumed. This inference can be obtained in a straightforward fashion from (33ii) (Zanuttini and Portner 2003).

Next, let us consider the listing sentence, repeated below (37), in which the scalar inference is more trivial:

- (37) Zhangsan bu-dan he-le xiangbin, hong-jiu, bai-jiu, he
 Zhangsan NEG-only drink-PFV champagne red-wine white-wine and
 pijiu/mijiu, lian [mijiu]_F/[pijiu]_F dou he-le bushao.
 beer/rice.wine include rice.wine/beer all drink-PFV some
 ‘Zhangsan drank champagne, red wine, white wine, beer/rice wine, and even some
 rice wine/beer.’

Under our analysis, the domain widening still applies to the *lian.dou* construction, but the reason why the scalar inference is trivialized is that it is the quantity information that is under discussion here (so the sentence is about how many types of alcohol Zhangsan drank, rather than what alcohol he drank) (Kay 1990). In this case, the domain widening works on a numeric scale, and the fact that Zhangsan drank 5 types of alcohol is always more informative than that he drank 4 types of alcohol, regardless of what kind of alcohol he drank.²⁰

Finally, our theory can account for sentences with both *shenzhi* and *lian.dou*:

- (38) *We are discussing Zhangsan’s tolerance of spicy food:*
 Zhangsan shenzhi lian habanero dou chi.
 Zhangsan even include habanero all eat
 ‘Zhangsan even eats habanero.’

In this case, *shenzhi* replaces the Empha.Assert operator as the clausal EVEN operator and compares alternative propositions in C. Note that *shenzhi* should not be considered as an overt realization of the Empha.Assert operator as they employ different scales. In (38), for example, *shenzhi* compares the worlds in which Zhangsan eats each pepper in D2 (including habanero) with those worlds in which Zhangsan eats each pepper in D1 (not including habanero), and the gradable property G could be about ‘Zhangsan’s tolerance of spicy food.’ The fact that Zhangsan also eats habanero indicates that he has a higher tolerance than the alternative situation where he eats other peppers, but not habanero.

²⁰ Under Zanuttini and Portner’s (2003) analysis, we might assume that the initial domain D1 consists of different situations, in which Zhangsan drank 1 type of wines, 2 types of wines, etc., and the widened domain D2 consists of every situation in D1, plus the situation where Zhangsan drank 5 types of wines. We thus obtain the inference that the added situation in D2 is more informative than those in D1.

3.3 Summary

We have demonstrated that there are two ways of obtaining scalar inferences. One is from the clausal EVEN *shenzhi*, which maps accessible worlds to degrees on a scale based on a contextually supplied gradable property (Greenberg 2018). The other is from an emphatic operator that induces domain widening, and the scalar inference involves the unlikelihood or the quantity scale. The *lian.dou* construction that consists of *lian* ‘include’ and *dou* ‘all’ overtly spells out the domain-widening process in Mandarin Chinese.

There are several advantages of our analysis. Semantically, we show that both the additive and scalar inferences can be directly derived from the inherent meanings of *lian* ‘include’ and *dou* ‘all’. Syntactically, our proposal accounts for the syntactic distributions of *shenzhi* and the *lian*-phrase. We correctly predict that the *lian*-phrase must occur to the left of *dou* because the *lian*-phrase forms a constituent with the unpronounced QP, which provides the initial resource domain associated with *dou*, and therefore, the *lian*-phrase has the same distribution as the QPs. Furthermore, our analysis echoes a version of Hole’s (2015) syntactic proposal that the focus association can be distributed over the entire clause. In particular, we propose that an EVEN clause is partitioned into three parts: (i) an evaluative phrase that expresses the speaker’s scalar evaluation of the clause, (ii) an Ad-Focus phrase, which marks the focused phrase and activates the scalar EVEN operator in the EvalP, and (iii) a background marking for the rest of the clause. In Mandarin Chinese, the clausal EVEN can be realized as *shenzhi* ‘even’ or as a covert Emph.Assert operator, and the *lian*-phrase occupies the Ad-Focus phrase that specifies the focused phrase associated with the scalar EVEN operator at the clausal level, and the *dou*-marked phrase provides the background part in the focus configuration. We assume that the *lian*-NP occupies the specifier position of the phrase headed by *dou*; however, our semantic analysis does not hinge upon this assumption:

- (39) [EvalP *shenzhi*/Emph.Assert [IP/AspP [Ad-FocusP *lian*-NP] [*dou* (background) ...]]]

We will return to this issue in Section 5 and see that this tripartite structure can be found cross-linguistically in the EVEN constructions in different languages.

3.4 A brief note on *lian.ye*

So far, we only discuss the additive scalar *lian.dou* sentences, but sometimes, the universal/distributive particle *dou* can be replaced by the additive particle *ye* ‘also’ without significantly changing the additive and scalar meanings of the sentence, as shown in (40):

- (40) Zhangsan *lian* *weijifen* *ye/dou* *na-dao* A.
 Zhangsan include calculus also/all get A
 ‘Zhangsan got an A even in calculus.’
 (Inferences: Zhangsan also got an A in other subjects, and it is most unlikely to get an A in calculus.)

Since *dou* ‘all’ and *ye* ‘also’ are not fully equivalent in other constructions (see below), this raises the question of why they can both occur with *lian* and give rise to the same additive scalar meanings.

It has been observed that *ye* and *dou* are interchangeable when it is associated with *wh*-indeterminates or *renhe*-NPs, Hole (2004) calls this the parametric use of *ye* ‘also’ and *dou* ‘all’:

- (41) a. Zhangsan shenme shuiguo ye/dou chi.
 Zhangsan what fruit also/all eat
 ‘Zhangsan eats every fruit.’
 b. Gang lai zheli de shihou, Lisi renhe-ren ye/dou bu-renshi.
 just arrive here DE time Lisi any-person also/all NEG-know
 ‘When he first came here, Lisi did not know anybody.’

Elaborating on Hole (2004), Yang (2018, 2020) argues that the parametric *ye* forces a scalar ordering among entities in the associated domain, while *dou* can be used more broadly (when there is an ordering or not). Yang’s main argument comes from free choice sentences involving *wulun/bulun* ‘no matter’:

- (42) a. Wulun an na-ge jian dou/?*ye keyi.
 no.matter press which key all/also can
 ‘No matter which key you press, it will work.’
 b. Buguan bai-tian wan-shang, ta dou/ye yao dian-zhe you-deng.
 no.matter day night 3SG all/also need light-PROG oil-lamp
 ‘No matter it is day or night, he keeps the oil lamp lit.’

In (42a) the domain of entities associated with *dou* is unordered or partially ordered (each key is equally likely to be pressed), making *ye* inadequate. In (42b), on the other hand, it is less likely that one would keep the lamp lit during the day, so there is a total ordering relation between the entities in the domain associated with *ye* (day <likely night). In such cases, *ye* can be used (as well as *dou*). We wish to point out that there seems to be speaker variation (as well as dialectal variation) regarding these judgments. For many native speakers of Mandarin Chinese, including ourselves, while *dou* is indeed preferred in (42a), the use of *ye* only slightly degrades the acceptability. Nonetheless, we think that the domain-widening analysis proposed here may provide a clue as to why the distinction between *dou* and *ye* is neutralized in *lian*-sentences. Just like *dou* ‘all’, *ye* ‘also’ is compatible with a QP that defines an initial resource domain (D1), such as a *wh*-indeterminate or a *renhe*-NP:

- (43) Zhangsan shenme shuiguo_{D1} ye/dou chi.
 Zhangsan what fruit also/all eat
 ‘Zhangsan eats every fruit.’

Therefore, both *ye* ‘also’ and *dou* ‘all’ can provide an initial resource domain for the domain-widening mechanism, and *lian* ‘include’ simply expands the initial domain by adding an entity to it. It does not matter whether the initial domain contains an ordered set or not. We contend that this is why the contrast between *dou* ‘all’ and *ye* ‘also’ is neutralized with *lian* ‘include’. Note that the account also applies to listing sentences. For example, the initial domain in (37) contains a totally ordered sequence, which is a numeric scale in terms of the quantities of drinks (1 type < 2 types < 3 types...), and *lian* simply adds an additional type of drink to the scale. Therefore, *dou* ‘all’ or *ye* ‘also’ can be used in this context.²¹

²¹ It should be noted that one can also use *ye dou* ‘also all’ together in many cases. For us, the use of *ye* and *dou* together seems equivalent to the individual use of either *ye* or *dou*. Furthermore, while Mandarin Chinese does not allow the reversed *dou ye* ‘all also’ sequence, this sequence can be found in Southern Min, which permits both ‘also all’ and ‘all also’ in free variation. We will leave the dialectal variation to our future study.

4 Scalarity and distributivity

The literature on *dou* generally focuses on the ambiguity of *dou* between its distributive use and its scalar use (there are also other uses of *dou*, but we will not discuss them here). In particular, when a universal QP is quantified by *dou*, only the distributive reading is available, but not the scalar reading, as in (44):

- (44) [Mei-ge/renhe ren]_i dou_i mai-le yi-ben shu.
 every-CLF/any person all buy-PFV one-CLF book
 a. ‘Everyone bought a book.’ [distributive]
 b. *‘It is the most unlikely that everyone bought a book.’ [scalar]

The apparent complementary distribution between the two uses of *dou* suggests that it might be possible to derive the distributive meaning of *dou* from its scalar meaning (or vice versa). We review two recent attempts here (Liu 2016, 2017; Y. Xiang 2016, 2020) and show that our analysis provides a simpler alternative.

Liu (2016, 2017) argues that *dou* is inherently a scalar EVEN, and its distributive meaning is obtained when the likelihood-based scalar relation is trivialized by another entailment relation associated with *dou*. Take (44) as an example. In its distributive use, a logical entailment relation holds among the alternatives to ‘everyone’. Namely, [everyone bought a book] entails that [most people bought a book], [some people bought a book], and [one person bought a book]. Liu argues that the logical entailment relation is stronger than the likelihood-based relation, and hence, the former trivializes the latter. Liu’s account, however, fails to notice that *lian* cannot be added to (44) (**Lian mei-ge ren...* ‘including everyone...’), so the reason why (44) does not have a scalar inference is that it does not trigger domain widening. In fact, (45) displays that both the distributive/universal meaning (from *dou*) and the scalar meaning (from EVEN...*lian*-NP) can be obtained since the distributive and the scalar meanings target different NPs:

- (45) [Cunzi-li de mei-ge-ren]_i, EVEN wo [lian gang-chusheng
 village-in DE every-CLF-person 1SG include just-born
 de baobao]_F dou_i renshi.
 DE infant all know
 ‘Among people in the village, I know everyone, including (even) newborn babies.’

In (45), the distributive relation holds between the distributive particle *dou* and the universal QP in the topic position (which also provides the initial resource domain associated with *dou*), and the scalar inference comes from domain widening, which includes newborn babies introduced by *lian* into the initial domain. The distributive meaning does not jeopardize the scalar meaning in any way, and both meanings contribute to the additive and scalar inferences of EVEN. The only notable difference between (44) and (45) is that when we have a *lian*-NP as the argument of the verb, it occupies the argument position and forces the universal QP to surface as a covert argument (or as a topic). However, this is a syntactic issue rather than a semantic one, so Liu’s semantic solution is not well-motivated.

From the opposite direction, Y. Xiang (2016, 2020) attempts to derive the scalar meaning of *dou* from its distributive/exhaustive meaning. Simply put, Xiang’s idea is that the distributive meaning comes from the exhaustive semantics of *dou* in selecting the strongest alternative, so that in the example like [John, Bill, and Mary *dou* left], the distributive alternative [John left and Bill left and Mary left] entails, or is logically stronger than, each of

the non-distributive sub-alternatives (such as [John and Bill left] or [John and Mary left]...). Therefore, *dou* always results in a distributive reading. In the case of *lian.dou* sentences, Xiang (2016, 2020) argues that the scalar reading also arises from *dou*'s being able to pick up the strongest alternative, but in this case, the most unlikely alternative (rather than the logically strongest one) is selected, and this generates the scalar reading of *dou*. Xiang's attempt is conceptually appealing, but it also suffers from the same problem as Liu's (2016) analysis. Namely, it wrongly predicts that the distributive reading and the scalar reading are in complementary distribution. However, as shown in (45), the two readings are not mutually incompatible, but instead, they simply mark different targets. Additionally, as acknowledged in Xiang (2020: footnote 26), it is not entirely clear why the shift from the logical strength to the likelihood is forced in the *lian.dou* construction. Xiang suggests that "one possibility is that the shift to likelihood is licensed by accenting the associate of *dou*." Our analysis, on the other hand, lays out the hidden mechanism behind such 'accenting' by assuming that an emphatic assertion operator associates with the *lian*-NP phrase, and this leads to domain widening, which introduces the strongest alternative into the initial domain.

5. Cross-linguistic implications

Using additive particles to express the scalar EVEN meaning is a cross-linguistically robust phenomenon. Such patterns can be found in Japanese *mo* 'also' (Nakanishi 2006; Kobuchi-Philip 2009), Hindi *bhii* 'also' (Lahiri 1998), Vietnamese *cả...cũng* 'also' (Hole 2017; Zimmerman 2017), Spanish *incluso* 'include' (Lahiri 2008), and across many languages in Europe, such as, German, Finnish, Swedish, Hungarian, Estonian, etc. (Schwarz 2005; Abrusán 2007; Gast and van der Auwera 2011). At the same time, many of these languages also employ a non-additive focus particle to express the scalar EVEN meaning, on a par with *shenzhi* in Mandarin Chinese. The analysis proposed in this paper suggests that we can analyze these patterns more systematically – if the EVEN meaning comes from a pure scalar particle (often in the form of a clausal adverbial), then it does not necessarily trigger an additive inference. On the other hand, if the EVEN meaning comes from an additive particle, then we obtain both additive and scalar inferences. In this section, we support our conclusion by examining the data patterns found in German, Vietnamese, and Persian.

In German, several lexical items may correspond to *even* in English. We concentrate on two of them, *sogar* and *auch nur* (Schwarz 2005). The former is a pure scalar particle (which has a clausal adverbial use), while the latter is a lexicalized combination of an additive particle *auch* 'also' plus *nur* 'only'.²² In addition, *auch nur* is a negative polarity item that can only be used in downward entailing contexts (presumably due to the presence of a scalar *nur* 'only/just', which creates a scale reversal effect; see Guerzoni 2003). Despite the fact that *auch nur* is an NPI, what is crucial to our analysis is that in terms of the availabilities of scalar and additive inferences, the distinction between *sogar* and *auch nur* in German is very much like the difference found in Mandarin *shenzhi* and *lian.dou*. Schwarz (2005) shows that

²² In addition to *sogar* and *auch nur*, the particles *selbst* '(lit.) self' and *einmal* '(lit.) once' may also carry the EVEN meaning (see Eckardt 2001 for an analysis of *selbst*). Due to the limited scope of our study, we set aside these two particles, but it might be interesting to note that both *selbst* and *einmal* give rise to both additive and scalar meanings. With respect to *selbst*, Eckardt (2001) argues that its EVEN meaning comes from association with the Emph.Assert operator (like *lian.dou* in Mandarin Chinese), so it could be argued that its additive meaning also arises from domain widening. We will leave a full comparison between the two particles to our future research. We thank an anonymous reviewer for bringing our attention to the German data. Another reviewer also points out that *auch noch* '(lit.) also still', which does not require NPIs, might be relevant here. However, we are currently unaware of any analysis of *auch noch* and will leave this topic for future research.

sogar can be a pure scalar particle that does not necessarily carry the additive inference, while the additive meaning of *auch nur* (aside from its scalar meaning) is obligatory. Therefore, *sogar* can be felicitously used in exclusive contexts, such as those in (46) (from Schwarz 2005: 141–142). These patterns suggest that *sogar* behaves more like *shenzhi* in Mandarin Chinese:

- (46) a. Hans hat sogar die SILBERMEDAILLE_F gewonnen.
 Hans has even the silver-medal won
 ‘Hans has even won the silver medal.’
 b. Hans ist sogar ein ASSOCIATE PROFESSOR_F in Linguistik.
 Hans is even an associate professor in linguistics
 ‘Hans is even an associate professor in linguistics.’

As for the additive *auch nur*, Schwarz (2005) argues that the additive meaning is obligatory, in addition to the scalar meaning (see also Guerzoni 2003). Consider the examples below:

- (47) a. Keiner von uns hat auch nur den ERSTEN_F Band gelesen.
 none of us has also only the first volume read
 ‘None of us has even read the first volume.’
 b. Keiner von uns hat auch nur die BRONZEMEDAILLE_F gewonnen.
 none of us has also only the bronze-medal won
 ‘None of us has even won the bronze medal.’

(47a) has a scalar inference in which the first volume is the most likely book to be read (and it asserts that none of us has read it), and at the same time, it implies that none of us has read the second or third volume, either (the additive inference), and the additive inference cannot be canceled. In the same vein, (47b) implies that none of us won silver or gold medals, either.

Auch nur therefore behaves like *lian.dou* in the sense that both require the additive inference to be present. In fact, Schwarz (2005) contends that the additive inference is part of the truth-conditional meaning of the *auch nur*, which is again consistent with our analysis that the additive meaning comes from the lexical meaning of *dou* (or *ye*). The distinction between *auch nur* and *lian.dou* is therefore limited to the presence of *nur* ‘only/just’, which requires that *auch nur* be used as a negative polarity item (see Guerzoni 2003 for a decompositional analysis of *auch nur*).

Another language that shares a very similar pattern with Mandarin Chinese is Vietnamese (we thank Tran N. Phan (p.c.) for providing us with the data and insights). In Vietnamese, there are at least three lexical items that can obtain the EVEN reading (see also Hole 2017 and Zimmerman 2017), *thậm chí* ‘even’, *đến...cũng* ‘(lit.) to...also’, and *cả...cũng* ‘also...also.’ The different EVENS can be used independently (48a, b) or co-occur in a single sentence (48c):

- (48) a. Nó thậm chí uống rượu_F.
 3SG even drink alcohol
 ‘He even drinks alcohol.’
 b. Nó đến/cả/đến cả rượu_F cũng uống.
 3SG to/also/to.also alcohol also drink
 ‘He even drinks alcohol.’

- c. Nó thậm chí đến cả rượu cũng uống.
 3SG even to also alcohol also drink
 ‘He even drinks alcohol.’

Our analysis paves the way for us to disentangle the hidden mechanisms behind different uses of EVENS in Vietnamese. Consider the exclusive context in (49) and the co-occurrence with universal QPs in (50). We see that the clausal adverbial EVEN *thậm chí* behaves like *shenzhi* in Mandarin Chinese (they are purely scalar) and the additive EVEN *cả...cũng* ‘also...also’ behaves like *lian.dou* in Mandarin Chinese (both additive and scalar):²³

(49) The gold-medal context: no additive inference

Although people thought Zhangsan would not perform well in the game, he actually won a medal:

- a. Hơn thế, nó thậm chí giành được [huy chương vàng]_F.
 moreover 3SG even fight gain medal gold
 ‘What’s more, he even took the gold medal.’
 b. #Hơn thế, nó cả [huy chương vàng]_F cũng giành được.
 moreover 3SG also medal gold also fight gain

(50) Co-occurrence with a universal QP: no additive inference

- a. Giáo sư thậm chí đánh rớt [mọi]_F sinh viên trong lớp.
 professor even fail every student in class
 ‘The professor even failed all students in the class.’
 b. *Giáo sư cả [mọi]_F sinh viên trong lớp cũng đánh rớt.
 professor also every student in class also fail

In Persian, Mortazavinia (2020) observes that there are two lexical items that may bring about the EVEN meaning, including the pure scalar particle *hattā* ‘even’ (similar to *shenzhi* in Chinese) and the additive particle *ham* ‘also/too’ (both form constituents with their adjacent NP/PP):

(51) Persian *hattā*: scalar (Mortazavinia 2020:152)

Claire is not a highly educated person and also never wanted to marry a highly educated person. But to everyone’s surprise, she married someone who teaches at university.

A: Did Claire marry an assistant professor?

B: *Hattā* bā ye ostadyār ezdevāj kard.

SCAL with a associate.professor marriage did

‘She *even* married an [associate professor]_F.’ (with the scalar inference, but not an additive one)

²³ Meanwhile, the particle *đến* ‘come.to’ in Vietnamese displays an interesting and more complicated pattern that behaves like the overt counterpart of the Emph.Assert operator in Mandarin Chinese (it cannot be used in exclusive contexts and employs a likelihood scale when it occurs with (*cả*)...*cũng* ‘also’). Due to the limit of space, we leave a detailed comparative study between EVENS in Mandarin Chinese and Vietnamese to our future study.

(52) Persian *ham*: additive (Mortazavinia 2020:154)

There was a swimming competition at school and medals were given to the students who reached the end of the pool before others. Kian and Kaveh were the first to reach the end at the same time. They both won the first title and were given two gold medals.

A: Tell me about the competition yesterday!

B: Kiān talā gereft. in dāfe, Kāve ham talā_F gereft.

Kian gold got this time, Kaveh too gold got.

‘Kian won a gold. This time, Kaveh *too* won a gold.’ (with the additive inference)

Mortazavinia (2020: 156) shows that *hattā* can occur with *ham*, and the *hattā*-XP-*ham* pattern triggers both scalar and additive inferences:

(53) Scalar-additive context in Persian

Amir is not a good friend and hasn’t been talking to us for many years. No one expected him to show up at last night’s party.

A: Last night’s party was very busy.

Hattā amir_F ham oomade bood.

even Amir too come was

‘Even Amir showed up.’ (with both scalar and additive inferences)

In terms of our proposal, we can analyze the Persian data by assuming that *hattā* is a pure scalar particle like *shenzhi* in Mandarin Chinese, even though *hattā* occurs at the phrasal level (as it is always attached to the focused NP/PP). This surface syntactic difference, however, can be analyzed away if we assume the *shenzhi* is phonetically realized in the focus operator position, while *hattā* is phonetically realized with the focused NP/PP, and it signals the association with a covert scalar EVEN operator. Our proposal is illustrated in (54). What is crucial here is that *shenzhi* and *hattā* are both scalar items, and they are not necessarily additive. On the other hand, the particle *ham* is like *lian.dou* in that they are inherently additive particles, and they acquire both additive and scalar inferences through focus association with a higher scalar EVEN operator, as illustrated in (55):

(54) a. M. Chinese: pure scalar

[*shenzhi*_{scalar}... [NP_F]]

b. Persian: scalar

[EVEN_{scalar}... [*hattā*-NP_F]]

(55) a. M. Chinese: both scalar and additive

[*shenzhi*_{scalar}/Emph.Assert... [*lian*_{additive}-NP] *dou*...]

b. Persian: scalar and additive

[EVEN_{scalar} ... [*hattā*-NP-*ham*_{additive}]...]

Finally, we return to *even* in English. Wagner (2015) argues that the NP-*even* always triggers an additive inference (i.e., when *even* locally modifies an NP), while the VP-*even* (where *even* modifies the whole clause in its domain) is not always additive (from Wagner 2015: 3):

(56) a. John married an Assistant Professor, and Bill even married an Associate Professor.

b. #John married an Assistant Professor, and Bill married even an Associate Professor (→ Odd because Bill seems to have married two people).

Yet, if our observation is on the right track, it cannot simply be a distinction between NP-*even* and VP-*even*, but whether the *even* corresponds to a clausal-level scalar particle (like *shenzhi*) or a phrasal-level additive particle (like *lian.dou*). In fact, an earlier example (15), repeated below as (57), represents a counterexample to Wagner’s generalization because *even* here associates with an adjacent subject NP, but the NP-*even* does not necessarily trigger an additive inference (because no one else ate thistles) (from Szabolcsi 2017):

- (57) *Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known for favoring thistles) takes a bite but spits it out.*

Those thistles must be really prickly! Even Eeyore_{REF} spat thistles out.

To solve the asymmetry of NP-*even* in (56) and (57), we propose that the particle *even* in (56) can only be analyzed as a phrasal EVEN because a clausal EVEN (being a speaker-oriented adverb) cannot be placed between a verb and its object (**John saw unexpectedly Bill*). Thus, the object NP-*even* in (56) must correspond to the phrasal EVEN (like *lian.dou* in Mandarin Chinese). On the other hand, a subject NP-*even* may, in principle, correspond to a clausal EVEN or a phrasal EVEN since a speaker-oriented adverb may occur in the clause-initial position as well as the clause-internal position (*Unfortunately, John has accepted* and *John unfortunately has accepted* are both acceptable; see Cinque 1999: 110). Yet, there is reason to believe that the NP-*even* in (57) corresponds to a clausal EVEN. According to Zhang (2022), the main contribution of *even* in (57) corresponds to a QUD about a degree question, such as “*How prickly are those thistles*” or “*Are those thistles prickly?*” This is exactly like the function of the clausal EVEN *shenzhi* in Mandarin Chinese (i.e., *shenzhi* measures a contextually salient gradable property G = the prickliness of those thistles), and therefore, (57) does not trigger an additive inference since it corresponds to a clausal EVEN. Note that in Mandarin Chinese, this type of example must occur with *shenzhi*, but not with *lian.dou*:

- (58) *Lisi is always the last person that comes to class, but today is so unusual that...*

Qita-ren hai-mei lai zhi-qian...
other-person yet-NEG come before
'Before anyone shows up...'
a. #Lian [Lisi]_F dou (zaojiu) yijing lai-le.
include Lisi all early already come-PFV
'Even Lisi has already arrived early.'
b. [Lisi]_F shenzhi (zaojiu) yijing lai-le.
Lisi even early already come-PFV
'It is even the case that Lisi has already arrived (very early).'

Summing up our discussion, we see that cross-linguistically, EVEN can be dissected into a scalar component and an additive component. The scalar component functions as a pure scalar particle without necessarily conveying an additive meaning (e.g., *shenzhi* in Mandarin Chinese). In contrast, the additive component (e.g., *lian.dou* in Mandarin Chinese) inherently carries an additive meaning, which may give rise to scalar interpretations through association with a higher scalar EVEN operator. This allows us to explain how different EVEN particles may bring about distinct scalar and additive inferences.

6. Conclusion

Our study suggests that there are two ways to obtain the EVEN meaning. The first approach involves merging a clausal adverbial EVEN (exemplified by *shenzhi* in Mandarin Chinese) in the speaker-oriented domain. The clausal EVEN triggers a scalar inference that expresses how the speaker compares the accessible possible worlds by mapping them to degrees on a scale of a contextually supplied gradable property (Greenberg 2018). In this case, additivity is a pragmatic implicature that arises from the accessibility condition (in normal/typical situations) so it can be canceled when the alternatives are mutually incompatible. The second approach, on the other hand, comes from an additive particle that is merged at the phrase level (exemplified by *lian.dou* in Mandarin Chinese). The additive phrase may associate with a higher scalar operator and give rise to the scalar meaning through domain widening. Our theory provides a novel account of the correlations between different instances of EVENS and the inferences they trigger. The study also lays the foundation for understanding the widely observed cross-linguistic pattern in which additive particles may convey scalar meanings. It is our hope that our findings will invite further cross-linguistic research on this topic.

Abbreviation

We follow the Leipzig Glossing Rules with the following addition: EXP = experiential aspect; SFP = sentence-final particle.

References

- Abrusán, Marta. 2007. *Even* and free choice *any* in Hungarian. *Proceedings of Sinn und Bedeutung* 11: 1–15.
- Badan, Linda, and Francesca Del Gobbo. 2015. The *even*-construction and the low periphery in Mandarin Chinese. In *The cartography of Chinese syntax*, ed. Wei-Tien Dylan Tsai, 33–74. Oxford: Oxford University Press.
- Bellert, Irena. 1977. On semantic and distributional properties of sentential adverbs. *Linguistic Inquiry* 8(2): 337–351.
- Chen, Zhuang, and Yael Greenberg. 2022. A novel argument for an *even*-like semantics of Mandarin *dou*. *SALT* 32: 751–771.
- Cheng, Lisa, and Rint Sybesma. 1999. Bare and not-so-bare nouns and the structure of NP. *Linguistic Inquiry* 30(4): 509–542.
- Chierchia, Gennaro. 2006. Broaden your views: Implicatures of domain widening and the "logicality" of language. *Linguistic Inquiry* 37(4): 535–590.
- Chierchia, Gennaro. 2013. *Logic in grammar: Polarity, free choice, and intervention*. Oxford: Oxford University Press.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. New York: Oxford University Press.
- Constant, Noah, and Chloe C. Gu. 2010. Mandarin 'even,' 'all' and the trigger of focus movement. *Proceedings of the 33rd Annual Penn Linguistics Colloquium* 16(1): 21–30.
- Crnič, Luka. 2011. *Getting even*. PhD dissertation. MIT.
- Drubig, H. Bernhard. 2003. Toward a typology of focus and focus construction. *Linguistics* 41(1): 1–50.
- Eckardt, Regine. 2001. Reanalysing *selbst*. *Natural Language Semantics* 9(4): 371–412.
- Ernst, Thomas. 2002. *The syntax of adjuncts*. Cambridge: Cambridge University Press.
- Ernst, Thomas. 2009. Speaker-oriented adverbs. *Natural Language & Linguistic Theory* 27(3): 497–544.
- Fauconnier, Gilles. 1976. *Etude de certains aspects logiques et grammaticaux de la quantification et de l'anaphore en français et en anglais*. Paris: Champion.
- Francis, Naomi. 2019. *Presuppositions in focus*. PhD dissertation. MIT.

- Gast, Volker, and Johan van der Auwera. 2011. Scalar additive operators in the languages of Europe. *Language* 87(1): 2–54.
- Greenberg, Yael. 2016. A novel problem for the likelihood-based semantics of *even*. *Semantics and Pragmatics* 9: 1–28.
- Greenberg, Yael. 2018. A revised, gradability-based semantics for *even*. *Natural Language Semantics* 26(1): 51–83.
- Grice, Paul. 1975. Logic and conversation. In *Speech acts*, eds. Peter Cole and Jerry L. Morgan, 41–58. New York: Academic Press.
- Guerzoni, Elena. 2003. Why even ask: On the pragmatics of questions and the semantics of answers. PhD dissertation. MIT.
- Haegeman, Liliane. 2012. *Adverbial clauses, main clause phenomena, and composition of the left periphery*. New York: Oxford University Press.
- Heim, Irene, and Angelika Kratzer. 1998. *Semantics in generative grammar*. Malden, MA: Blackwell.
- Herburger, Elena. 2000. *What counts: Focus and quantification*. Cambridge, MA: MIT Press.
- Hole, Daniel. 2004. *Focus and background marking in Mandarin Chinese: System and theory behind cai, jiu, dou and ye*. New York: Routledge.
- Hole, Daniel. 2015. A distributed syntax for evaluative ‘only’ sentences. *Zeitschrift für Sprachwissenschaft* 34(1): 43–77.
- Hole, Daniel. 2017. A crosslinguistic syntax of scalar and non-scalar focus particle sentences: The view from Vietnamese and Chinese. *Journal of East Asian Linguistics* 26(4): 389–409.
- Horn, Larry. 1969. A presuppositional approach to *only* and *even*. *Chicago Linguistic Society (CLS)* 5: 98–107.
- Iatridou, Sabine, and Sergei Tatevosov. 2016. Our *even*. *Linguistics and Philosophy* 39(4): 295–331.
- Kadmon, Nirit, and Fred Landman. 1993. Any. *Linguistics and Philosophy* 16(4): 353–422.
- Karttunen, Lauri, and Stanley Peters. 1979. Conventional implicature. In *Syntax and semantics, Vol. 11: Presupposition*, eds. Choon-Kyu Oh and David A. Dinneen, 1–56. New York: Academic Press.
- Kay, Paul. 1990. Even. *Linguistics and Philosophy* 13(1): 59–111.
- Kobuchi-Philip, Mana. 2009. Japanese *mo* ‘also/even’ and *shika* ‘except for/only’. *Proceedings of Sinn und Bedeutung* 14: 219–236.
- Kratzer, Angelika. 1981. The notional category of modality. In *Words, worlds, and contexts*, eds. H. J. Eikmeyer and H. Rieser, 38–74. New York: de Gruyter.
- Kratzer, Angelika. and Junko Shimoyama. 2002. Indeterminate pronouns: The view from Japanese. In *Proceedings of the Third Tokyo Conference on Psycholinguistics*, ed. Yukio Otsu, 1–25. Tokyo: Hituzi Syobo.
- Krifka, Manfred. 1991. A compositional semantics for multiple focus constructions. *SALT* 1: 127–158.
- Krifka, Manfred. 1995a. The semantics and pragmatics of polarity items. *Linguistic Analysis* 25(3–4): 209–257.
- Krifka, Manfred. 1995b. Common nouns: a contrastive analysis of Chinese and English. In *The generic book*, eds. G. N. Carlson & F. J. Pelletier, 398–411. Chicago: The University of Chicago Press.
- Krifka, Manfred. 1998a. Additive particles under stress. *SALT* 8: 111–128.
- Krifka, M. 1998b. Scope inversion under the rise-fall contour in German. *Linguistic Inquiry* 29: 75–112.
- Krifka, Manfred. 2006. Association with focus phrases. In *The architecture of focus*, eds. Valéria Molnar and Susanne Winkler, 105–136. Berlin: Mouton de Gruyter.

- Lahiri, Utpal. 1998. Focus and negative polarity in Hindi. *Natural Language Semantics* 6(1): 57–123.
- Lahiri, Utpal. 2008. The semantics and pragmatics of some scalar expressions in Spanish. *International Journal of Basque Linguistics and Philology (ASJU)* 42(2): 359–389.
- Lin, Jowang. 1998. Distributivity in Chinese and its implications. *Natural Language Semantics* 6(2): 201–243.
- Liu, Mingming. 2016. Varieties of alternatives. PhD dissertation. Rutgers University.
- Liu, Mingming. 2017. Varieties of alternatives: Mandarin focus particles. *Linguistics and Philosophy* 40(1): 61–95.
- Liu, Mingya. 2012. *Multidimensional semantics of evaluative adverbs*. Leiden: Brill.
- Mayol, Laia, and Elena Castroviejo. 2013. (Non)integrated evaluative adverbs in questions: A cross-Romance study. *Language* 89(2): 195–230.
- Mortazavinia, Marzieh. 2020. Second language acquisition of the semantics of focus-sensitive operators in Persian. In *The Routledge handbook of second language acquisition and pedagogy of Persian*, ed. Pounesh Shabani-Jadidi, 147–179. London: Routledge.
- Nakanishi, Kimiko. 2006. *Even, only, and negative polarity in Japanese*. *SALT* 16: 138–155.
- Potts, Christopher. 2005. *The logic of conventional implicatures*. Oxford: Oxford University Press.
- Rooth, Mats. 1985. *Association with focus*. PhD dissertation. University of Massachusetts, Amherst.
- Rooth, Mats. 1992. A theory of focus interpretation. *Natural Language Semantics* 1(1): 75–116.
- Rullmann, Hotze. 1997. *Even, polarity, and scope*. In *Papers in experimental and theoretical linguistics*, eds. Martha Gibson, Grace Wiebe, and Gary Libben, 40–64. Edmonton, Canada: University of Alberta.
- Schwarz, Bernhard. 2005. Scalar additive particles in negative contexts. *Natural Language Semantics* 13(2): 125–168.
- Shank, Scott. 2004. Domain widening. PhD dissertation. The University of British Columbia.
- Shyu, Shu-ing. 2004. (A)symmetries between Mandarin Chinese *lian...dou* and *shenchi*. *Journal of Chinese Linguistics* 32(1): 71–128.
- Shyu, Shu-ing. 2016. Minimizers and *even*. *Linguistics* 54(6): 1355–1395.
- Shyu, Shu-ing. 2018. The scope of *EVEN*: Evidence from Mandarin Chinese. *Language and Linguistics* 19(1): 156–195.
- Szabolcsi, Anna. 2017. Additive presupposition are derived through activating focus alternatives. *Proceedings of the Amsterdam Colloquium* 21: 455–464.
- Thomason, Richmond H., and Robert Stalnaker. 1973. A semantic theory of adverbs. *Linguistic Inquiry* 4(2), 195–220.
- von Stechow, Kai. 1994. Restrictions on quantifier domains. PhD dissertation. University of Massachusetts.
- Wagner, Michael. 2015. *Even and the syntax of additivity*. Paper presented at Linguistics Colloquium, University of Chicago.
- Wilkinson, Karina. 1996. The scope of *even*. *Natural Language Semantics* 4(3): 193–215.
- Winter, Yoad. 2016. *Elements of formal semantics: An introduction to the mathematical theory of meaning in natural language*. Edinburgh: Edinburgh University Press.
- Wold, Dag. E. 1996. Long distance selective binding: The case of focus. *SALT* 6: 311–328.
- Xiang, Ming. 2008. Plurality, maximality and scalar inferences: A case study of Mandarin *dou*. *Journal of East Asian Linguistics* 17(3): 227–245.
- Xiang, Yimei. 2016. Interpreting questions with non-exhaustive answers. PhD dissertation. Harvard University.

- Xiang, Yimei. 2020. Function alternations of the Mandarin particle *dou*: Distributor, free choice licensor, and 'even'. *Journal of Semantics* 37(2): 171–217.
- Yang, Zhaole. 2018. Mandarin *yě* and scalarity. *Studies in Chinese Linguistics* 39(2): 155–178.
- Yang, Zhaole. 2020. *Yě, yě, yě: On the syntax and semantics of Mandarin yě*. Amsterdam: LOT. Retrieved from <https://hdl.handle.net/1887/123042>
- Zanuttini, Raffaella, and Paul Portner. 2003. Exclamative clauses: At the syntax-semantics interface. *Language* 79(1): 39–81.
- Zhang, Linmin. 2022. The presupposition of *even*. *SALT* 32: 249–269.
- Zimmermann, Malte. 2017. Scalar particles and contrastive topics in English and Vietnamese. In *MIT working papers in linguistics: Proceedings of IATL 31*, ed. Elitzur A. Bar-Asher Siegal, 123–152. Cambridge, MA: MIT Press.